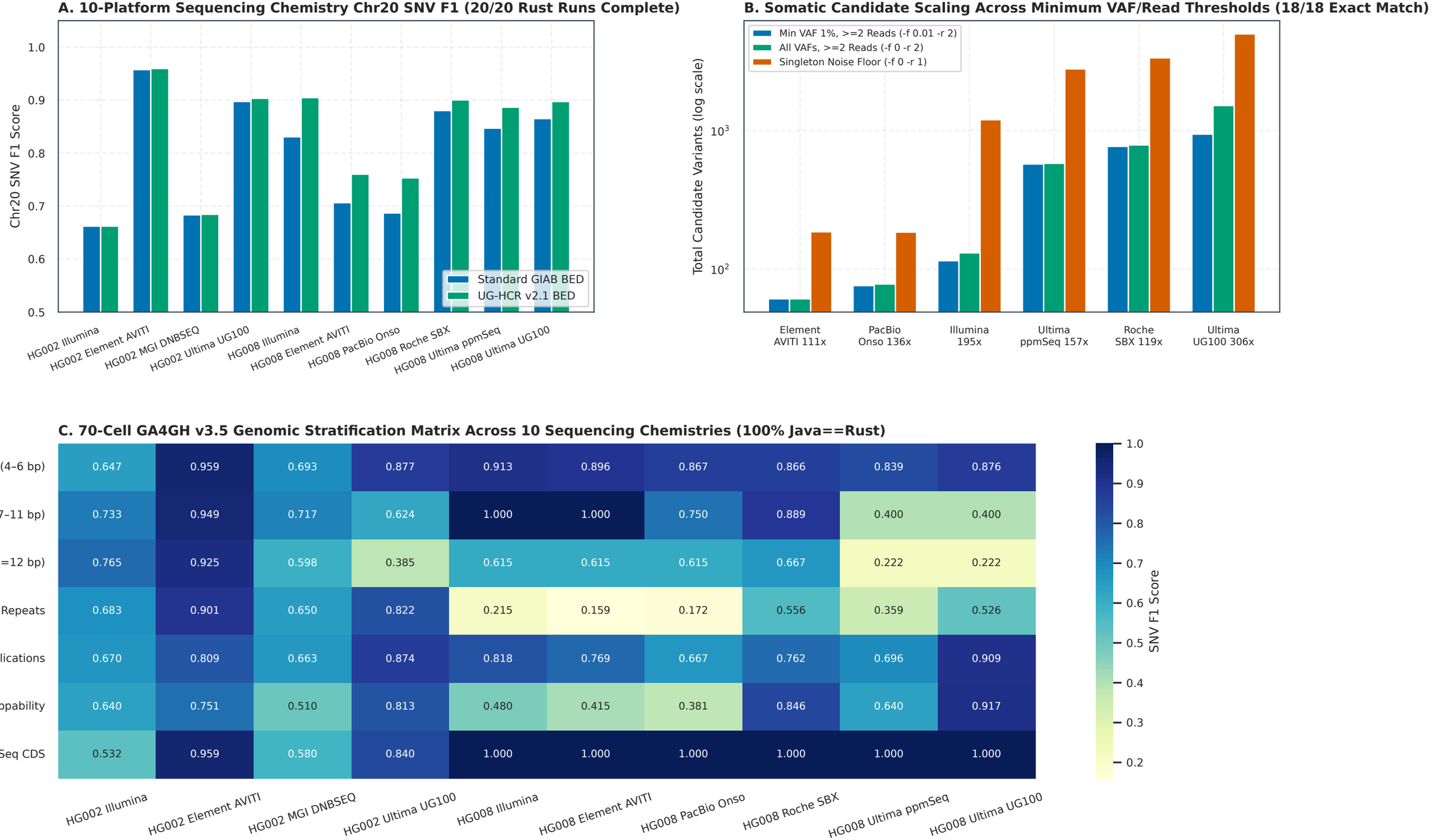
A. 10-Platform Sequencing Chemistry Chr20 SNV F1 (20/20 Rust Runs Complete)
Chr20 SNV F1 Score
1.0
0.9
0.8
0.7
0.6
0.5
Standard GIAB BED
UG-HCR v2.1 BED
HG002 Illumina
HG002 Element AVITI
HG002 MGI DNBSEQ
HG002 Ultima UG100
HG008 Illumina
HG008 Element AVITI
HG008 PacBio Onso
HG008 Roche SBX
HG008 Ultima ppmSeq
HG008 Ultima UG100
B. Somatic Candidate Scaling Across Minimum VAF/Read Thresholds (18/18 Exact Match)
Total Candidate Variants (log scale)
10^3
10^2
Min VAF 1%, >=2 Reads (-f 0.01 -r 2)
All VAFs, >=2 Reads (-f 0 -r 2)
Singleton Noise Floor (-f 0 -r 1)
Element AVITI 111x
PacBio Onso 136x
Illumina 195x
Ultima ppmSeq 157x
Roche SBX 119x
Ultima UG100 306x
C. 70-Cell GA4GH v3.5 Genomic Stratification Matrix Across 10 Sequencing Chemistries (100% Java==Rust)
Homopolymer (4–6 bp) 0.647 0.959 0.693 0.877 0.913 0.896 0.867 0.866 0.839 0.876
Homopolymer (7–11 bp) 0.733 0.949 0.717 0.624 1.000 1.000 0.750 0.889 0.400 0.400
Homopolymer (>=12 bp) 0.765 0.925 0.598 0.385 0.615 0.615 0.615 0.667 0.222 0.222
Tandem Repeats 0.683 0.901 0.650 0.822 0.215 0.159 0.172 0.556 0.359 0.526
Segmental Duplications 0.670 0.809 0.663 0.874 0.818 0.769 0.667 0.762 0.696 0.909
Low Mappability 0.640 0.751 0.510 0.813 0.480 0.415 0.381 0.846 0.640 0.917
RefSeq CDS 0.532 0.959 0.580 0.840 1.000 1.000 1.000 1.000 1.000 1.000
HG002 Illumina
HG002 Element AVITI
HG002 MGI DNBSEQ
HG002 Ultima UG100
HG008 Illumina
HG008 Element AVITI
HG008 PacBio Onso
HG008 Roche SBX
HG008 Ultima ppmSeq
HG008 Ultima UG100
SNV F1 Score
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2